

S1

```
class s1 {  
    public static void main(String args[]) {  
        if(args.length != 2) {  
            System.out.println("Please enter 2 numbers");  
            return;  
        }  
        int a = Integer.parseInt(args[0]);  
        int b = Integer.parseInt(args[1]);  
  
        System.out.println("Sum = " + (a + b));  
  
        if(a > b)  
            System.out.println("Largest = " + a);  
        else  
            System.out.println("Largest = " + b);  
  
        if(a % 2 == 0)  
            System.out.println(a + " is Even");  
        else  
            System.out.println(a + " is Odd");  
  
        if(b % 2 == 0)  
            System.out.println(b + " is Even");  
        else  
            System.out.println(b + " is Odd");  
    }  
}
```

S2

```
import java.util.Scanner;
```

```
// Employee Class
```

```
class Employee {
```

```
    int empld;
```

```
    String name;
```

```
    double salary;
```

```
    String dob;
```

```
void acceptEmployee(Scanner sc) {
```

```
    System.out.print("Enter Employee ID: ");
```

```
    empld = sc.nextInt();
```

```
    sc.nextLine();
```

```
    System.out.print("Enter Employee Name: ");
```

```
    name = sc.nextLine();
```

```
    System.out.print("Enter date of birth: ");
```

```
    dob = sc.nextLine();
```

```
    System.out.print("Enter Salary: ");
```

```
    salary = sc.nextDouble();
```

```
}
```

```
void displayEmployee() {
```

```
    System.out.println("\nEmployee Details");
```

```
    System.out.println("-----");
```

```
    System.out.println("Employee ID : " + empld);
```

```
    System.out.println("Employee Name : " + name);
```

```
    System.out.println("date of birth : " + dob);
```

```
    System.out.println("Salary : " + salary);
```

```
}
```

```
}
```

```
// Manager Class inherits Employee
```

```
class Manager extends Employee {
```

```
    String department;
```

```
    void acceptManager(Scanner sc) {
```

```
        acceptEmployee(sc);
```

```
        sc.nextLine();
```

```
        System.out.print("Enter Department: ");
```

```
        department = sc.nextLine();
```

```
    }
```

```
    void displayManager() {
```

```
        System.out.println("\nManager Details");
```

```
        System.out.println("-----");
```

```
        System.out.println("Employee ID : " + empId);
```

```
        System.out.println("Employee Name : " + name);
```

```
        System.out.println("Salary : " + salary);
```

```
        System.out.println("Department : " + department);
```

```
    }
```

```
}
```

```
// SalesManager Class inherits Manager
```

```
class SalesManager extends Manager {
```

```
    double commission;
```

```
    void acceptSalesManager(Scanner sc) {
```

```
        acceptManager(sc);
```

```
System.out.print("Enter Commission: ");  
commission = sc.nextDouble();  
}
```

```
void displaySalesManager() {  
    System.out.println("\nSales Manager Details");  
    System.out.println("-----");  
    System.out.println("Employee ID : " + empId);  
    System.out.println("Employee Name : " + name);  
    System.out.println("Salary : " + salary);  
    System.out.println("Department : " + department);  
    System.out.println("Commission : " + commission);  
    System.out.println("Total Salary : " + (salary + commission));  
}  
}
```

```
// Main Class
```

```
public class s2 {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
        int choice;  
  
        do {  
            System.out.println("\n===== MENU =====");  
            System.out.println("1. Employee");  
            System.out.println("2. Manager");  
            System.out.println("3. Sales Manager");  
            System.out.println("4. Exit");  
            System.out.print("Enter your choice: ");
```

```
choice = sc.nextInt();
```

```
switch (choice) {
```

```
    case 1:
```

```
        Employee e = new Employee();
```

```
        e.acceptEmployee(sc);
```

```
        e.displayEmployee();
```

```
        break;
```

```
    case 2:
```

```
        Manager m = new Manager();
```

```
        m.acceptManager(sc);
```

```
        m.displayManager();
```

```
        break;
```

```
    case 3:
```

```
        SalesManager sm = new SalesManager();
```

```
        sm.acceptSalesManager(sc);
```

```
        sm.displaySalesManager();
```

```
        break;
```

```
    case 4:
```

```
        System.out.println("Exiting Program...");
```

```
        break;
```

```
    default:
```

```
        System.out.println("Invalid Choice!");
```

```
}
```

```
} while (choice != 4);
```

```
        sc.close();
    }
}
```

S3

//Emp.java

```
package employeee;

import java.util.Scanner;

public class emp
{
    int empId;

    String empName, category;

    double basic, hra, da, allowance, pf, incomeTax, gross, net;

    public void getData()
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");

        empId = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Employee Name: ");

        empName = sc.nextLine();

        System.out.print("Enter Employee Category: ");

        category = sc.nextLine();

        System.out.print("Enter Basic Pay: ");

        basic = sc.nextDouble();
    }

    public void calculate()
    {
        hra = basic * 0.20;
```

```

        da = basic * 0.50;

        allowance = basic * 0.10;

        pf = basic * 0.12;

        incomeTax = basic * 0.05;


        gross = basic + hra + da + allowance;

        net = gross - (pf + incomeTax);
    }

    public void display()
    {
        System.out.println("Employee ID    : " + empld);
        System.out.println("Employee Name  : " + empName);
        System.out.println("Employee Category : " + category);
        System.out.println("Basic Pay    : " + basic);
        System.out.println("HRA        : " + hra);
        System.out.println("DA         : " + da);
        System.out.println("Allowance   : " + allowance);
        System.out.println("PF         : " + pf);
        System.out.println("Income Tax  : " + incomeTax);
        System.out.println("Gross Pay   : " + gross);
        System.out.println("Net Pay     : " + net);
    }
}

```

S3.java

```

import employee.emp;

public class s3
{
    public static void main(String args[])
    {
        emp e = new emp();

        e.getData();
    }
}

```

```
        e.calculate();  
        e.display();  
    }  
}
```

S4

```
import java.util.Scanner;  
  
class InvalidMarksException extends Exception {  
    public InvalidMarksException(String message) {  
        super(message);  
    }  
}  
  
public class s4 {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int rollNo;  
        int marks;  
        try {  
            System.out.print("Enter Roll Number: ");  
            rollNo = sc.nextInt();  
            System.out.print("Enter Marks: ");  
            marks = sc.nextInt();  
            if(marks > 100) {  
                throw new InvalidMarksException("Marks out of Bound");  
            }  
            System.out.println("\nStudent Details");  
            System.out.println("Roll Number : " + rollNo);  
            System.out.println("Marks : " + marks);  
        } catch(InvalidMarksException e) {
```



```
break;
```

case 2:

```
Scanner f1 = new Scanner(new File("student.txt"));
```

```
while(f1.hasNextLine())
```

```
    System.out.println(f1.nextLine());
```

```
f1.close();
```

```
break;
```

case 3:

```
Scanner f2 = new Scanner(new File("student.txt"));
```

```
System.out.print("Enter Roll No to Search: ");
```

```
int s = sc.nextInt();
```

```
boolean found = false;
```

```
while(f2.hasNextLine())
```

```
{
```

```
    String data = f2.nextLine();
```

```
    if(data.startsWith(s + " "))
```

```
    {
```

```
        System.out.println(data);
```

```
        found = true;
```

```
    }
```

```
}
```

```
if(!found)
```

```
    System.out.println("Student Not Found");
```

```
f2.close();
```

```
break;
```

case 4:

```
System.out.println("Program Ended");
```

```
break;
```

default:

```
System.out.println("Invalid Choice");
```

```
}
```

```
    } while(ch != 4);
```

```
}
```

```
}
```

S6

```
import javax.swing.*.*;
```

```
import java.awt.event.*;
```

```
public class s6 extends JFrame implements ActionListener
```

```
{
```

```
    JLabel l1,l2,l3,l4,l5;
```

```
    JTextField t1,t2;
```

```
    JTextArea ta;
```

```
    JRadioButton m,f;
```

```
    JComboBox cb;
```

```
    JButton b1,b2;
```

```
s6()
```

```
{
```

```
    setTitle("Admission Form");
```

```
    setSize(400,450);
```

```
    setLayout(null);
```

```
    l1=new JLabel("Name");
```

```
    l2=new JLabel("Gender");
```

```
    l3=new JLabel("Course");
```

```
    l4=new JLabel("Address");
```

```
    l5=new JLabel("Contact");
```

```
    l1.setBounds(50,50,100,25);
```

```
    l2.setBounds(50,100,100,25);
```

```
l3.setBounds(50,150,100,25);
```

```
l4.setBounds(50,200,100,25);
```

```
l5.setBounds(50,290,100,25);
```

```
add(l1); add(l2); add(l3); add(l4); add(l5);
```

```
t1=new JTextField();
```

```
t2=new JTextField();
```

```
ta=new JTextArea();
```

```
t1.setBounds(150,50,150,25);
```

```
ta.setBounds(150,200,150,60);
```

```
t2.setBounds(150,290,150,25);
```

```
add(t1); add(ta); add(t2);
```

```
m=new JRadioButton("Male");
```

```
f=new JRadioButton("Female");
```

```
m.setBounds(150,100,80,25);
```

```
f.setBounds(230,100,80,25);
```

```
ButtonGroup bg=new ButtonGroup();
```

```
bg.add(m);
```

```
bg.add(f);
```

```
add(m); add(f);
```

```
String s[]{"BCA","BBA","MBA","MCA"};
```

```
cb=new JComboBox(s);
```

```

cb.setBounds(150,150,150,25);
add(cb);

b1=new JButton("Submit");
b2=new JButton("Reset");

b1.setBounds(80,350,100,30);
b2.setBounds(200,350,100,30);

add(b1); add(b2);

b1.addActionListener(this);
b2.addActionListener(this);

setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setVisible(true);
}

public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==b1)
    {
        JOptionPane.showMessageDialog(this,
            "Name: "+t1.getText()+
            "\nGender: "+(m.isSelected()?"Male":"Female")+
            "\nCourse: "+cb.getSelectedItem()+
            "\nAddress: "+ta.getText()+
            "\nContact: "+t2.getText());
    }

    if(e.getSource()==b2)

```

```

{
    t1.setText("");
    t2.setText("");
    ta.setText("");
    m.setSelected(false);
    f.setSelected(false);
    cb.setSelectedIndex(0);
}
}

```

```

public static void main(String args[])
{
    new s6();
}
}

```

S7

//Server.java

```

import java.io.*;
import java.net.*;

```

```

public class server {
    public static void main(String[] args) {
        try {
            ServerSocket serverSocket = new ServerSocket(5000);
            System.out.println("Server is waiting for client...");

            Socket socket = serverSocket.accept();
            System.out.println("Client connected");

            BufferedReader in = new BufferedReader(

```

```

        new InputStreamReader(socket.getInputStream()));

    PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

    String message = in.readLine();
    System.out.println("Message from client: " + message);

    out.println(message);

    socket.close();
    serverSocket.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}
}

//Client.java
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class client {
    public static void main(String[] args) {
        try {
            Socket socket = new Socket("localhost", 5000);

            BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));

            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter message: ");
String msg = sc.nextLine();

out.println(msg);
String response = in.readLine();
System.out.println("Echo from server: " + response);
socket.close();
} catch (Exception e) {
    System.out.println(e);
}
}
}
```